

Deep learning segmentation of soil constituents in 3D X-ray CT images

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ABSTRACT

Accurate segmentation of soil constituents in X-ray CT imagery is critical for advancing our understanding of soil structure dynamics. However, difficulties arise because of the overlapping X-ray attenuation of soil constituents, which makes segmentation based on voxel intensity alone impossible. In this study, we explore the potential of nnUNet, a deep learning-based semantic segmentation model, for applications in soil science. We evaluated nnUNet on three challenging datasets: (1) complex soil structure with numerous material classes with overlapping gray value ranges, (2) fine roots in noisy images, and (3) permafrost with gradual gray value transitions between sediment types. The performance of nnUNet was compared to other reference methods, namely Ilastik, Rootline v.2, and manual thresholding. The Dice scores indicated overall good model performance across all three datasets. Compared to the reference methods, nnUNet outperformed Ilastik on Dataset 1. For Dataset 2, nnUNet produced segmentations with fewer false-positive roots than Rootline v2; however, this came at the expense of missing fine roots that were barely visible to the naked eye and, therefore, not annotated. In Dataset 3, we encountered challenges in annotating classes due to gradual transitions in voxel intensity. Our study underscores that deep learning models like nnUNet perform well for the segmentation of complex soil structures and could assist the development of a generalized segmentation model, thereby fostering standardization in soil structure analysis. To achieve this, increased access to diverse and well-curated annotations is still necessary.

1. Introduction

X-ray CT is a powerful technique for 3D nondestructive imagery of soil samples. The images resulting from typical X-ray CT scans are usually grayscale images, for which the intensity of the gray value is directly related to the density of the internal features of the sample. Due to this ability to visualize the internal structure of opaque objects, X-ray CT opens many promising avenues for the study of soil structure dynamics and its feedback with biotic factors such as root growth (Phalempin et al., 2022) and bioturbation (Capowiez et al., 2011), and abiotic factors such as freeze/thaw cycles (Leuther and Schlüter, 2021) or tillage operation (Schlüter et al., 2020). Deepening our knowledge on the structure and dynamics of soil constituents constitutes a step towards a holistic characterization of soil properties and their link with soil functions, which is particularly relevant, for example, in the context of carbon sequestration in soils (Rabot et al.,

2018). X-ray CT also enables to repetitively scan soil pots with growing roots. Therefore, it constitutes an excellent tool for the study of root growth in soil (Lippold et al., 2021), which is instrumental for the phenotyping of beneficial root traits in the context of sustainable agriculture.

Semantic segmentation is a computer vision task in which the goal is to categorize each pixel in an image into a class. It is a necessary step to quantitatively derive relevant metrics from X-ray CT images, e.g., the connectivity and tortuosity of the pore network or the length and volume of the root system. Semantic segmentation of X-ray CT images of soil samples presents a significant challenge due to the inherent complexity of soil structure and its constituents. Undisturbed soil samples usually contain numerous constituents of interest, which greatly vary in size and shape (e.g., soil matrix, rocks, living roots, degrading roots, root-induced pores, earthworm burrows, desiccation

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cracks, etc.). These constituents can span only one voxel in width (for example, very fine roots) or several hundreds of voxels (e.g., an earthworm burrow). In addition, classes such as roots, plant litter, water-filled pores, and the soil matrix with subresolution porosity share overlapping gray value ranges. This makes segmentation based solely on the voxel intensity error prone (Mooney et al., 2012). In addition to these challenges, segmentation of soil constituents is particularly difficult due to class imbalance (e.g., a lot of voxels comprise the soil matrix, but fewer voxels comprise the living roots).

Although X-ray CT is now fully established in various disciplines of geosciences (Withers et al., 2021), the processing of the resulting images still lacks standardization of methods. Global or locally adaptive thresholding are segmentation methods based on the definition of gray value thresholds, usually assessed with a histogram analysis based on an entire image (global) or on subimages or pixel neighborhoods (local). Due to their inherent simplicity, these methods can handle only limited complexity of data. They are prone to classification errors, notably because of the presence of partial volume voxels at the border between features and because of the background noise inherent to every image acquisition technique. These segmentation techniques are also usually not fully automated, that is, they require fine-tuning and adaptation for each dataset, sometimes even for each sample within a dataset. The definition of thresholds is also observer dependent, making it difficult to generalize it in a broader segmentation context (Baveye et al., 2010).

In recent years, machine learning-based segmentation methods have been developed and now offer a solution to the limitations of simpler thresholding methods. These methods exploit the potential of multidimensional image features to harness the information underlying in texture (e.g., the standard deviation of gray value within a given window of interest) and edges (e.g., the n^{th} derivative of gray value within a given window of interest). Thanks to their design principles, machine learning-based methods are well suited not only to harness the attenuation property of soil constituents but also to leverage their morphological characteristics for more accurate segmentation. For example, despite the similar X-ray attenuation of living roots and plant litter, these two can potentially be differentiated because living roots are typically elongated, whereas plant litter may take on a variety of shapes. Similarly, while both living and degrading roots are elongated, degrading roots tend to exhibit greater porosity and wrinkling, which is reflected in a higher standard deviation of gray values. Root channels and earthworm burrows are generally air-filled and cylindrical, but differ in size and texture, as burrows are often larger and partially filled with castings. These detailed distinctions highlight the potential of machine learning models to handle the complexity of soil structure. However, it remains to be tested how well these models perform in practice for the segmentation of such intricate and variable structures.

Machine learning models are usually classified into so-called “shallow” or “deep” learning models. Shallow learning methods such as WEKA (Arganda-Carreras et al., 2017), Ilastik (Berg et al., 2019) and Labkit (Arzt et al., 2022) have gained popularity in recent years, thanks to user-friendly software packages with a graphical user interface for class annotation. These tools typically use random forest classification for semantic segmentation, requiring users to manually define feature extraction parameters (i.e., the target feature and the sigma value of the Gaussian blur at which they are computed). Consequently, the quality of the segmentation is heavily dependent on the selection of appropriate features and parameters for each specific task. Although shallow models can perform well with smaller datasets thanks to their simplicity and reduced risk of overfitting, they often struggle with complex and larger datasets (Valente et al., 2023).

To deal with more complex datasets, deep learning models hold significant promises. Convolutional Neural Networks (CNNs), a class of deep learning models, have proven extremely efficient for image processing tasks by improving the accuracy, speed, and depth of image analysis (Litjens et al., 2017). These CNNs excel in recognizing patterns

and features in complex datasets, making them particularly well suited for analyzing X-ray CT scans. Although some efforts have been made to initiate the adoption of deep learning methods for image processing in soil sciences, most current methods have been developed to analyze root growth in 2D images obtained in controlled systems such as (mini-)rhizotron (Smith et al., 2020) or trenches (Teramoto and Uga, 2020). Fewer authors have focused on the processing of X-ray CT images. Notably, some deep learning algorithms have been developed for the detection of roots (Soltaninejad et al., 2020; Teramoto and Uga, 2024; Cassity et al., 2024), pores (Rippner et al., 2022; Wieland et al., 2021; Lavrukhin et al., 2021), and particulate organic matter (POM) (Rippner et al., 2022; Valdes-Korovkin et al., 2024) in soil samples. It should be noted that existing deep learning models for 3D X-ray CT image analysis have usually been developed for one specific segmentation task (i.e., three or four classes of interest, at most), which makes their applicability and transferability to other datasets questionable.

In this work, our aim is to broaden the adoption of deep learning methods for semantic segmentation of 3D X-ray CT images by the soil science community. To do so, we formulated four main objectives: (1) demonstrate the applicability and versatility of nnUNet, a deep learning-based semantic segmentation method initially developed for biomedical image analysis (Isensee et al., 2021), for the processing of 3D X-ray CT images of soil samples, (2) provide an efficient labeling strategy to obtain ground-truth annotations used as training datasets, (3) compare the performance of nnUNet against other existing segmentation methods, and (4) provide an open access resource with step by step instructions on how to transition from a 3D grayscale X-ray CT image to a prediction by nnUNet. To achieve these objectives, we applied nnUNet to three different datasets, each presenting unique segmentation challenges. Dataset 1 corresponds to soil samples extracted from typical agrosystems under different land uses (cropland versus grassland). Dataset 2 corresponds to soil columns filled with finely sieved soil used for root growth in the laboratory. The Dataset 3 corresponds to soil cores collected from permafrost. For each dataset, we used complimentary ground-truth data obtained through laboratory analysis after destructive sampling, which provided external validation for one or more constituents. We also compared the results of nnUNet with previous segmentation results published in the study of Schlüter et al. (2022) for Dataset 1, of Phalempin et al. (2021) for Dataset 2, and of Nitzbon et al. (2022) for Dataset 3. Based on our experience in using nnUNet and our expertise in image processing, we assessed the performance and versatility of nnUNet for applications in soil sciences.

2. Materials and methods

2.1. Dataset description

The first dataset originates from the study of Schlüter et al. (2022). The main challenge associated with this dataset is the presence of complex heterogeneous structures. Undisturbed soil samples were collected within the framework of the Global Change Experimental Facility (Schädler et al., 2019). They were extracted at a depth between 3 and 10 cm in field plots under five different farming practices, i.e., conventional farming, organic farming, intensively managed grassland cut by mowing, extensively managed grassland cut by mowing, and extensively managed grassland used as sheep pasture. These farming practices had profound impacts on soil characteristics, which manifested themselves in very contrasted pore morphologies and amount of POM (Schlüter et al., 2022). With this dataset, we tested the ability of nnUNet to segment numerous material classes sharing overlapping grayvalue ranges, having different sizes and shapes, and suffering from class imbalance.

The second dataset originates from the study of Phalempin et al. (2021). It represents the benchmark dataset of the “worse case” scenario, already thoroughly described in the aforementioned reference. The main challenge associated with the second dataset is posed by the

Table 1

Number of images per dataset, origin and dimensions of the samples as well as resolution at which the samples were scanned with X-ray CT.

Dataset	Sample origin	Soil structure	Total number of images in dataset	Sample height [cm]	Sample diameter [cm]	Voxel size [μm]
1	Agrosystems	Intact	75	4	5	30
2	Laboratory	Remoulded	12	4	7	45
3	Permafrost	Intact	32	0.9 to 1.9	7.5	50

inherent trade-off between pot size and image resolution. For X-ray imaging of intact roots in soil, it is desirable to grow plants in pots as large as possible to not artificially affect the architecture of the root system, but small enough to allow for an image resolution compatible with the recovery of the finest roots (Lucas and Vetterlein, 2022). The result of this trade-off is that fine roots have a diameter that is close to the image resolution. To acquire this dataset, maize plants (*Zea mays* L.) were previously in sieved soil and scanned with a rather low signal-to-noise ratio (i.e., a low number of projections per scan, resulting in faster scans). With this dataset, we tested the ability of nnUNet to retrieve all fine maize roots in the growing pots.

The third dataset originates from the study of Nitzbon et al. (2022), which aimed to unravel the structural heterogeneity of permafrost with X-ray CT. To do so, a core was sampled from an ice-rich Yedoma permafrost upland plateau on Kurungnakh Island, Lena River delta in north-eastern Siberia. For sampling, a soil pit was excavated at a maximum thaw depth of 20 cm, followed by drilling and coring the frozen material underneath to a depth of 164 cm. The diameter of the frozen core was 7.5 cm. The depths ranging from 20 to 80 cm were considered for the segmentation task, as the soil at those depths contained two types of sediment with ice inclusions. The primary challenge of Dataset 3 lies in the gradual transitions in gray values between the two types of sediment (an organic rich and a mineral rich one), as voxels represent a mix of sediments, at the resolution used. With this dataset, we tested the ability of nnUNet to properly segment the transitions between these organic-rich and mineral-rich classes.

For all three datasets, the samples were scanned with an industrial X-ray CT scanner (XT H 225, Nikon Metrology). The radiographs obtained were reconstructed into 3D tomograms with an isotropic voxel side length that varied according to the size of the samples. The scanner settings were adjusted for each dataset and are described in their respective references. The 8-bit grayscale tomographs were reconstructed using the CT Pro 3D software (Nikon Metrology). The grayscale range of each CT scan was normalized by setting the darkest and brightest 0.2 percentile to 0 and 255, respectively. Table 1 collects meta-information related to the number of images per dataset, the origin and dimension of the samples, and the resolution at which the samples were scanned with X-ray CT.

2.2. Collection of external ground-truth data

All three datasets have in common that external ground-truth data are available for at least one or more constituents of interest. The collection of external ground-truth was acquired through destructive sampling after X-ray CT scanning.

For Dataset 1, undisturbed soil cores were wet-sieved on a mesh of 630 μm to separate inorganic and organic constituents. For this dataset, we isolated the POM class, which consisted of living roots, degrading roots, and other types of POM such as litter, straws and leaf fragments. The collected mass of POM after sieving was determined after drying for 48 h at 70 °C and the POM content in a core was expressed after normalization per dry soil mass.

For Dataset 2, the root system was the class of interest. To collect the maize roots, the soil layers in the growing pots were isolated and the roots in those layers were washed off over a sieve having a 1 mm mesh size. The roots remaining on the mesh of the sieve were scanned with a flatbed scanner (EPSON perfection V700) and the obtained images were analyzed with WinRHIZO Pro™ (Version 2019a, Regent Instruments,

Canada) to calculate the root length. The results were then expressed in terms of root length density (RLD), after normalization by the volume of the sampled soil layer.

For Dataset 3, the targeted classes were the ice, organic matter, and mineral classes. To calculate the volumetric fractions of these classes, the volume and weight of all frozen subsamples were measured before the cores were allowed to thaw and dry in an oven at a temperature of 100 °C for 24 h. Knowing the weight before and after thawing and drying and making some assumptions on the density of each class, the volume fraction of each class of permafrost cores was calculated. For more information on these calculations, the reader is referred to Nitzbon et al. (2022).

2.3. Segmentation with reference methods

In addition to confronting the results of nnUNet with external ground-truth, we also compared the results with other segmentation methods. We considered these methods to be reference, as they have been successfully applied in previous studies of Schlüter et al. (2020), Phalempin et al. (2021), and Nitzbon et al. (2022).

For Dataset 1, the machine learning-based method “Ilastik” (Berg et al., 2019) was used to segment the data and was considered as the reference. Ilastik is an open source and user-friendly software tool for interactive image analysis. It allows users to train machine learning-based image classifiers according to sparse annotations fed by the user, and to directly apply these classifiers as batch jobs within ImageJ. In the reference study of Schlüter et al. (2020), four classes were distinguished according to visual assessment of images, i.e., air-filled pores, POM, soil matrix, and rocks. Random forest classification (“shallow learning”) was chosen and feature extraction was performed considering the original gray values as well as gradient and texture information after Gaussian smoothing with a strength of $\sigma = [0.3, 0.7, 1.0]$. The Ilastik classifier was trained with manually drawn annotations for each material class on a small number of images (6 out of 75). Before segmentation, the input images were downsampled by a factor of 2 to make image processing tractable. From the segmented images, the volume of POM was calculated by summing the number of voxels belonging to the POM class and the results were normalized per unit volume of soil.

For Dataset 2, we considered the root segmentation method “Rootline v.2” as the reference method. Rootline v.2 is a semi-automated deterministic root segmentation method that takes advantage of the cylindrical shape of roots and their connectivity as the main features. In the form of a FIJI/ImageJ macro, Rootline v.2 applies a series of image processing steps to grayscale input images (see figure 1 in Phalempin et al., 2021) in order to obtain binarized output images, i.e., images where roots are assigned to foreground and all other features of the image are assigned to background. Rootline v.2 automatically calculates the length of roots and relies on the BoneJ plugin (Doubé et al., 2010) to do so.

For Dataset 3, the global thresholding method for multiclass images, here referred to as “manual thresholding” (MT), was considered as the reference method. In this case, thresholds were carefully and tediously defined for each material class and adjusted for differences in illumination and proportion of material classes in each individual subcores (Nitzbon et al., 2022).

2.4. Segmentation with nnUNet

2.4.1. Class identification

The first step prior to segmentation with nnUNet was to identify the classes of interest. For Dataset 1, we identified nine classes based on visual inspection of the images (Fig. 1). For the mineral phase, we differentiated the soil matrix and denser particles such as sand grains, small rocks or concretions. For the pore phase, we identified pores created by abiotic factors such as tillage, wetting/drying, or freeze/thaw cycles and pores created by biotic factors (biopores) such as root decay or bioturbation. To differentiate abiotic pores from biopores, we defined biopores as those with an elliptical outline (cylindrical in 3D) and for which the surrounding filling material (either soil matrix or earthworm casts) occupied less than 50% of the pore space. The biopore class was further subdivided into two classes: the root-induced empty channels left after the decay of roots and the earthworm burrows created after the passage of earthworms. The differentiation between root-induced empty channels and earthworm burrows was mostly based on criteria such as size, shape, and texture. Unlike root-induced empty channels, earthworm burrows are generally much larger, do not have laterals that extend into the surrounding soil, and are occasionally filled with castings. For the organic phase, we distinguished the living roots and the degrading roots, the latter being root-like objects whose gray values were lower than those of the living roots (possibly because of decay or desiccation). We also identified the other POM class which we considered to be POM that is not root-derived (e.g., seeds, straw pieces, litter or leaf fragments). Finally, the aluminum wall of the soil cores was also identified. Visual examples of these classes are depicted in Fig. 1a to i.

For Dataset 2, we identified four distinct classes, that is, the roots, the polyethylene wall of the pot, the soil matrix, and the air-filled pores. These classes are depicted in Fig. 1m to r. For Dataset 3, we also identified four classes, namely ice, air-filled pores, organic matter, and mineral phase. Our definition of the ice class followed that of Nitzbon et al. (2022). We considered ice to be excess ice, that is, ice that is not retained by capillary forces after thawing. We also considered organic matter and mineral phase to be sediments A and B, respectively, as defined in Nitzbon et al. (2022). These two types of sediment contained characteristic proportions of ice, organic, and mineral matter (see table 1 in Nitzbon et al., 2022). These classes are shown at the Fig. 1s to x.

2.4.2. Image normalization

The second necessary step for image segmentation with nnUNet was to normalize the images. Image normalization was necessary to ensure that constituents had a consistent range of gray values within a dataset. Inconsistencies between the gray values of the constituents arose due to the percentile method used during reconstruction, which optimized the contrast by stretching the histogram. This contrast optimization was necessary, but also caused constituents to be assigned different gray values, if the volume proportion of pores and mineral phases in the images varied too much within a dataset. For instance, a constituent of lower inherent X-ray attenuation (e.g., ice) was assigned a greater gray value if a class of greater inherent X-ray attenuation (e.g., mineral soil) was absent in the image.

The images were normalized on the basis of the characteristic material of homogeneous and invariant X-ray attenuation. For Dataset 1, we made sure that the aluminum wall of the cores had a mean gray value of 245. For Dataset 2, we normalized the images so that the polyethylene wall of the columns had a mean gray value of 90. For Dataset 3, there was no wall that we could use as a reference. In that case, we chose the excess ice and the mineral phase as reference materials. We normalized the images so that the gray value peaks of the excess ice and mineral matter classes were assigned to 95 and 200, respectively.

Table 2

Number of cores randomly and purposefully selected as images for model training as well as the mean percentage of the images that was effectively annotated, for each dataset, with the standard deviation shown after the symbol “ $\pm$ ”.

Dataset	Training images	Randomly selected cores	Purposefully selected cores	Mean percentage of the training data annotated [%]
1	20	15	5	0.26 (± 0.004)
2	13	13	0	0.21 (± 0.01)
3	18	18	0	0.19 (± 0.02)

2.4.3. Annotation strategy

The third step necessary for image segmentation with nnUNet was to generate ground-truth annotation masks, which were then used as training volumes. To select training volumes, cores from each dataset were selected with a random number generator. For Dataset 1, some other cores were deliberately selected to also include features of interest which do not systematically appear in every image of the dataset, e.g., earthworm burrows. From the selected cores, we selected random slices (z_i) within substacks of 300 slices in the z -direction and performed dense 2D annotations in the middle slice (i.e., at $z_i + 150$) (Fig. 2). To do the annotations, we used the Labkit plugin (Arzt et al., 2022) implemented in FIJI/ImageJ (Schindelin et al., 2012). For Dataset 1, we segmented the soil matrix using the Otsu method (Otsu, 1979), which was bounded by a second threshold set at 220 to exclude rocks. We used the Otsu method for the middle slice because it allowed us to easily and quickly identify soil particles of all sizes. Then, manual annotations were done to correct for the classes that were improperly caught by the Otsu method, for instance, the living roots and POM. After this, living roots, POM, rocks, wall, earthworm burrows, and empty root channels were manually annotated. Finally, the abiotic pores were annotated using the “Bucket Tool”, which was practical to fill the regions bordered by the soil matrix. We spent one to two hours annotating the middle slice of each image, depending on the structure complexity. Approximately 90% of the middle slice was annotated. Once the middle slice was densely annotated, we spent another hour manually annotating salient features of interest in the slices adjacent to the middle slice. These features included crop residues, large concretions, large earthworm burrows, Enchytraeid cast, etc. For Datasets 2 and 3, the same procedure was employed but adjusted for their relevant classes, respectively.

This annotation strategy allowed us to minimize annotation efforts while still ensuring that all relevant classes were captured. Table 2 summarizes the number of cores randomly and purposefully selected, as well as the percentage of training data that was annotated, for each dataset. The percentage of the training data annotated was calculated by counting the number of annotated voxels over the total number of voxels, expressed in percentage. Fig. 1 illustrates the dense and sparse annotations for the relevant classes of each dataset.

2.4.4. nnUNet pipeline

Dataset fingerprint. nnUNet starts with the extraction of the fingerprint from the dataset based on a set of heuristic rules. For more information on these heuristic rules, the reader is referred to Isensee et al. (2021). During fingerprint extraction, the patch size is estimated based on the image properties. The patch size is a key hyperparameter that defines the side length of the subvolumes that the network processes during training. The estimation of the patch size is based on properties such as the image dimensions of the input image, the voxel size, and the memory constraints of the GPU (assumed to be 8 GB by default by nnUNet). Because the input images had similar properties (e.g., in terms of dimension and voxel size) across datasets, the resulting patch size was equal to 48, for all three datasets.

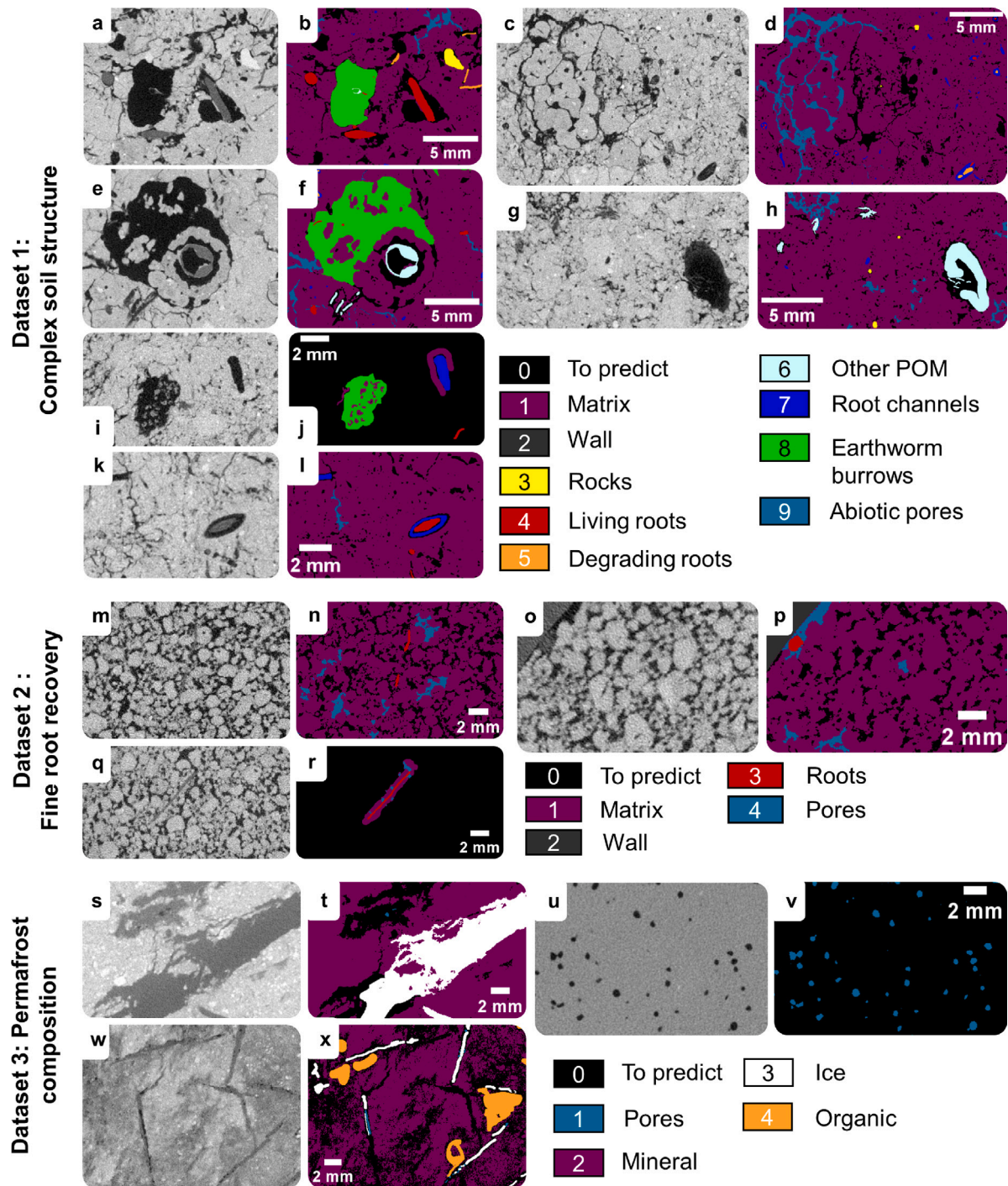


Fig. 1. Annotated classes for Dataset 1 (subfigure a to i), Dataset 2 (subfigure m to r) and Dataset 3 (s to x).

Image cropping. Given the sparse and local nature of the annotations, some parts of the training volumes did not contain useful information and were removed to improve computational efficiency. This was done by cropping the images along the z -axis. During cropping, special care was taken to ensure a buffer zone equivalent to half the patch size beyond the last annotated voxel in the z -direction. This was done to guarantee optimal sampling during training, as well as to capture sufficient context inside the cropped regions.

Training. The training was consistent with the standards of nnUNet and involved extensive data augmentation, including transformations such as rotations, scaling, Gaussian noise and blur, brightness adjustments, low simulated resolution, gamma corrections, and mirroring.

The model was trained for 1000 epochs, with each epoch consisting of 250 iterations, using a combined loss function that sums Dice and Cross-Entropy losses. This formulation implies that the lower the loss value, the better the model performance. The stochastic gradient descent was used as an optimizer, with a momentum of 0.99 and an initial learning rate of $1e^{-2}$, which progressively decayed according to a polynomial schedule (polyLR) with an exponent of 0.9. An ignore label was included in the training data to handle unlabeled voxels.

The only major deviation from the defaults of nnUNet was involved in the sampling strategy. Default sampling resulted in predictions focused solely on the middle due to the concentration of annotations in that slice. This was addressed by adding a random offset (ranging from

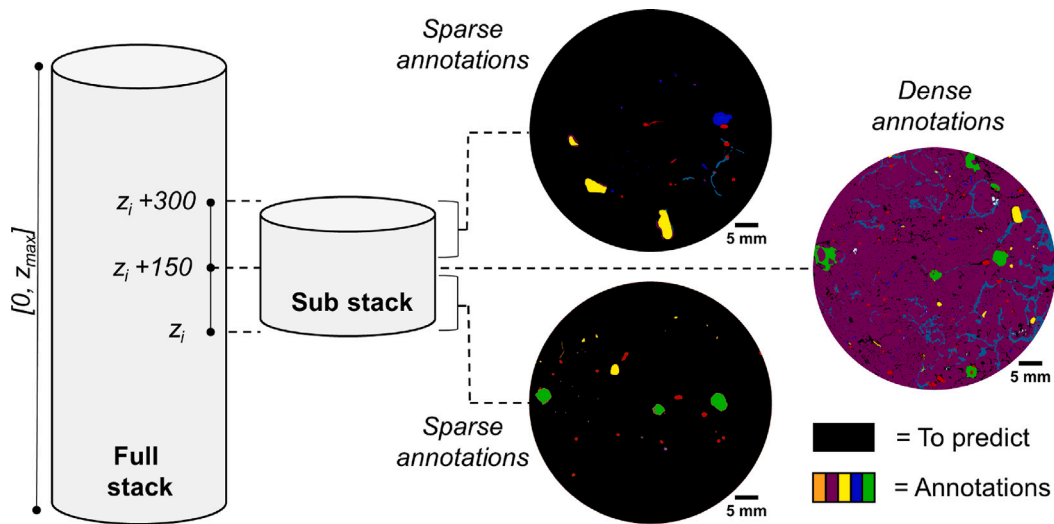


Fig. 2. Annotation strategy. Substacks were randomly selected to be part of the training datasets. The middle slices of these substacks were annotated densely class whereas slices adjacent to the middle slice were annotated sparsely for underrepresented features of interest.

0 to half the patch size) in each dimension to ensure more even spatial coverage during training.

Cross-validation. Model training was performed using a five-fold cross-validation scheme implemented by default in nnUNet. In this approach, the training data was partitioned into five equally (or nearly equally) sized subsets. For each fold, four fifths of the dataset were then used for training, while the remaining fifth was used for validation. This process was repeated five times while ensuring that each image was seen at least once during training and once during validation. To prevent data leakage and ensure the integrity of the evaluation process, crops of the same image were assigned to the same set when splitting the data into training and validation sets. This ensured a meaningful validation with results that accurately reflected the model's generalization capabilities. The Dice score was used as a performance metric (Eq. (1)).

$$\text{Dice} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \quad (1)$$

where TP is the number of correctly predicted positive voxels (the overlap between the predicted and actual positive regions), FP is the number of predicted positives that are actually negatives (regions predicted as positive but are false) and FN is the number of actual positives that are predicted as negatives (regions that should have been positive but were missed). The Dice score ranges between 0 and 1, where 0 means no overlap between the predicted and true regions and 1 indicates perfect overlap between the predicted and true regions.

Inference. Inference was performed using an ensemble of the five models generated during training. The models were ensembled by averaging softmax probabilities. The training folds and predictions were carried out on a high-performance computing cluster equipped with eight GPU nodes, each of them hosting one NVIDIA Tesla A100 80G GPU accelerator, which could be requested simultaneously via a job scheduling system. To run the predictions, the input images were split into ten substacks which had the same x-y dimensions but only a tenth of the z-extent, compared to the original input images. This allowed reducing the image size to < 500 MB, making sure that each substack fitted into the memory of the GPU. It also allowed to distribute the prediction jobs to the GPU nodes more efficiently. During the split of the original images, an overlap was considered to avoid edge artifacts. The overlap value was set equal to half of the patch size.

2.5. Assessment of segmentation performance

For Dataset 1, the segmentation performance of nnUNet and the Ilastik reference method was evaluated by plotting the total volume of

POM (i.e., the sum of living roots, the degrading roots and other POM classes) obtained from X-ray CT images against the gravimetric POM content obtained after wet-sieving (WS). Performance was assessed by fitting a linear regression to the pairs of points and retrieving the regression parameters. The intercept of the regression is indicative of the number of false positives in the predictions (i.e., the POM volume that would be falsely detected even if the sample did not contain any). The coefficient of determination (R^2) can be interpreted as the consistency of the error. A high R^2 value suggests a high consistency whereas a low R^2 value suggests a low consistency. The slope of the regression is a dimensionless scalar that is indicative of the POM density.

For Dataset 2, the segmentation performance of nnUNet and the reference method Routine v.2 was assessed by plotting the root length recovered after root segmentation against the root length recovered after wet-sieving. The average root recovery was calculated by fitting a linear regression to the pairs of points obtained with CT and after wet-sieving. The slope of the linear regression indicates the average root length recovery. A slope value greater than 1 suggests a majority of false positives over false negatives (oversegmentation), whereas a value smaller than 1 suggests that some roots were missed by the segmentation algorithm (presence of false negatives). Here also, the coefficient of determination of the regression indicates the error consistency.

For Dataset 3, the segmentation performance of nnUNet and the manual thresholding reference method was also assessed by plotting the volumetric fractions of ice, organic matter and the mineral phase against the volumetric fractions obtained after thawing and drying the samples. Similarly to Dataset 1 and 2, the performance of the segmentation methods was assessed by fitting linear regressions to the pairs of points and interpreting their parameters.

3. Results

3.1. Training

The time necessary to compute one epoch was approximately 40 s, which resulted in a total duration of approximately 12 h per training fold. This duration was roughly the same for all folds and datasets. The evolution of the loss function values averaged over the five folds is shown in Fig. 3. For all three datasets, the loss function starts at its highest values at the beginning of training and drastically decreases before stabilizing after approximately 150 epochs. For Dataset 2 and 3 in particular, the gains in model performance after 150 epochs were only marginal.

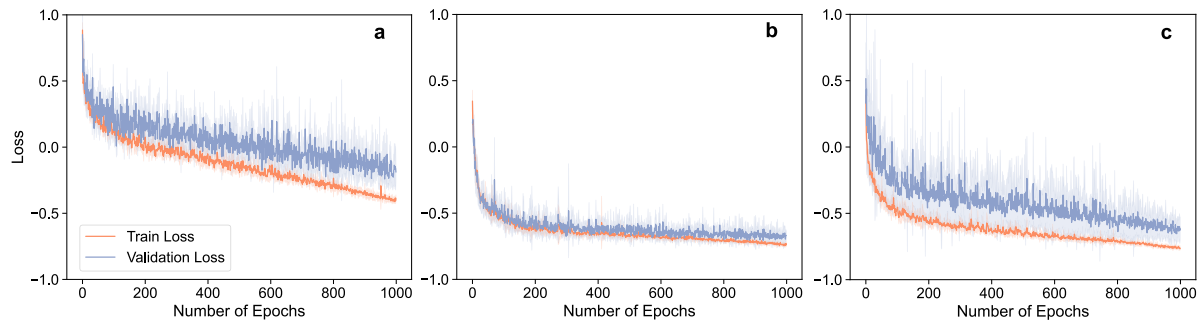


Fig. 3. Evolution of the averaged loss function during training (orange line) and validation (blue line) for Dataset 1 (subfigure *a*), Dataset 2 (subfigure *b*) and Dataset 3 (subfigure *c*). The shaded bands denote the standard deviation around the mean ($n = 5$).

Table 3

Metrics on the number of false positives (FP), false negatives (FN), true positives (TP), and the number of annotated voxels obtained after image annotation. The Dice scores for each class is also given after model validation for Dataset 1, 2 and 3.

Dataset 1	FP	FN	TP	Annotated voxels	Dice score
Matrix	44 741	59 984	355 490	415 474	0.87
Wall	2283	1485	496 569	498 054	0.99
Rocks	11 252	53 196	266 801	319 997	0.89
Living roots	40 131	56 271	325 443	381 714	0.87
Degrading roots	47 849	47 609	43 964	91 573	0.48
POM	49 858	63 789	230 235	86 812	0.80
Root channels	35 359	82 489	97 827	180 316	0.62
Earthworm burrows	82 072	52 626	695 622	748 248	0.91
Abiotic pores	89 122	122 406	687 473	809 879	0.86
Dataset 2					
Matrix	13 661	48 616	544 509	593 125	0.94
Wall	11 357	9374	1 195 066	1 204 440	0.99
Roots	15 449	126 725	351 853	478 578	0.83
Pores	14 177	9754	86 612	96 366	0.87
Dataset 3					
Pores	45 279	43 668	1 537 597	1 581 265	0.97
Mineral	36 800	31 402	1 605 363	1 636 765	0.98
Ice	45 085	50 759	2 370 318	2 421 077	0.98
Organic	53 951	48 845	637 117	685 962	0.92

The numbers of false positives, false negatives, true positives, the number of annotated voxels and Dice scores for each dataset are tabulated in Table 3. For Dataset 1, the Dice scores ranged from 0.48 to 0.99 and the ranking of the Dice scores for each individual class in increasing order was: degrading roots, root channels, other POM, abiotic pores, soil matrix, living roots, rocks, earthworm burrows, and wall. For Dataset 2, the Dice scores ranged from 0.83 to 0.99 and the ranking of the Dice scores for each individual class in increasing order was: living roots, abiotic pores, soil matrix and wall. For Dataset 3, the Dice scores ranged from 0.92 to 0.98 and the ranking of the Dice scores for each individual class in increasing order was: organic matter, pores, mineral and ice.

3.2. Segmentation performance

3.2.1. Dataset 1: Complex soil structure

nnUNet performed better than the Ilastik reference method for the intercept and R^2 values (Fig. 4a). For Ilastik, the intercept was equal to 7.9 mm³, while it amounted to 3.4 mm³ of POM per cm³ of soil for nnUNet. This corresponds to an average reduction in false positives by a factor of 2.3. The coefficient of determination with nnUNet reached 0.66, which also indicates higher reliability and consistency with nnUNet, compared to Ilastik ($R^2 = 0.53$). The gain in performance of nnUNet against Ilastik could also be visually assessed on the segmented volumes shown in Fig. 4b, and c. For the results obtained with Ilastik, many false positives were present and individual POM fragments could barely be distinguished. For the same samples

segmented with nnUNet, false positives were considerably reduced, and the cylindrical structure of the roots (living and degrading) and the planar structure of leaf fragments could easily be distinguished.

3.2.2. Dataset 2: Recovery of fine roots

The slope of the linear regression for the Rootine v.2 dataset exceeded that of nnUNet, and indicated that 63% of roots could be recovered with Rootine v.2 compared to 46% with nnUNet (Fig. 5a). Despite a lower net root recovery by nnUNet of 17%, the coefficient of determination suggested that the error was more consistent with nnUNet ($R^2 = 0.92$ for nnUNet against $R^2 = 0.81$ for Rootine v.2). The intercept value was roughly the same for nnUNet and Rootine v.2, suggesting a similar number of false positives for both methods. The difference in segmentation performance could be compared with two end members in terms of the root length they contain. For these samples *b* and *c*, the Fig. 5b and c show segmented cuboids, for which the roots captured by both nnUNet and Rootine v.2 are shown in black, the roots captured by Rootine v.2 only are shown in yellow, while the roots captured by nnUNet only are shown in turquoise. For the cuboid with a large amount of roots (sample *b*), Rootine v.2 captured more roots than nnUNet. For the cuboid with a low amount of roots (sample *c*), nnUNet and Rootine v.2 captured a similar number of roots (Fig. 5c).

3.2.3. Dataset 3: Permafrost constitution

The recovery of target classes with nnUNet and MT, compared to volume fractions recovered after thawing of the samples, is shown in Fig. 6a for the ice class, 6b for the mineral phase, and 6c for the organic matter class. A similar trend and spread of the data points was observed between MT and nnUNet, suggesting equally good results for both methods. However, the parameters recovered from the linear regressions (Table 4) indicated a deterioration of recovery (indicated by the slope, parameter *a*) and an increased presence of false positives (indicated by the intercept, parameter *b*) with nnUNet. Visual inspection of the segmentation results is also shown in Fig. 6d, e, f. In these figures, it is noticeable that more accurate and visually coherent segmentation results were achieved with nnUNet. Indeed, nnUNet reduced artifacts, such as misassignments caused by partial volume effects, leading to smoother and more consistent transitions between classes. However, with MT, the edges between the mineral class and the excess ice were always contoured by the organic matter class, which was obviously a thresholding artifact.

4. Discussions

4.1. Segmentation performance of nnUNet

For Dataset 1, the lowest Dice scores were obtained for the degrading roots (0.48) and the root-induced empty channels (0.62). This was not surprising, as these classes share similar features. In fact, both are elongated, tubular and had a similar diameter range. Also, both

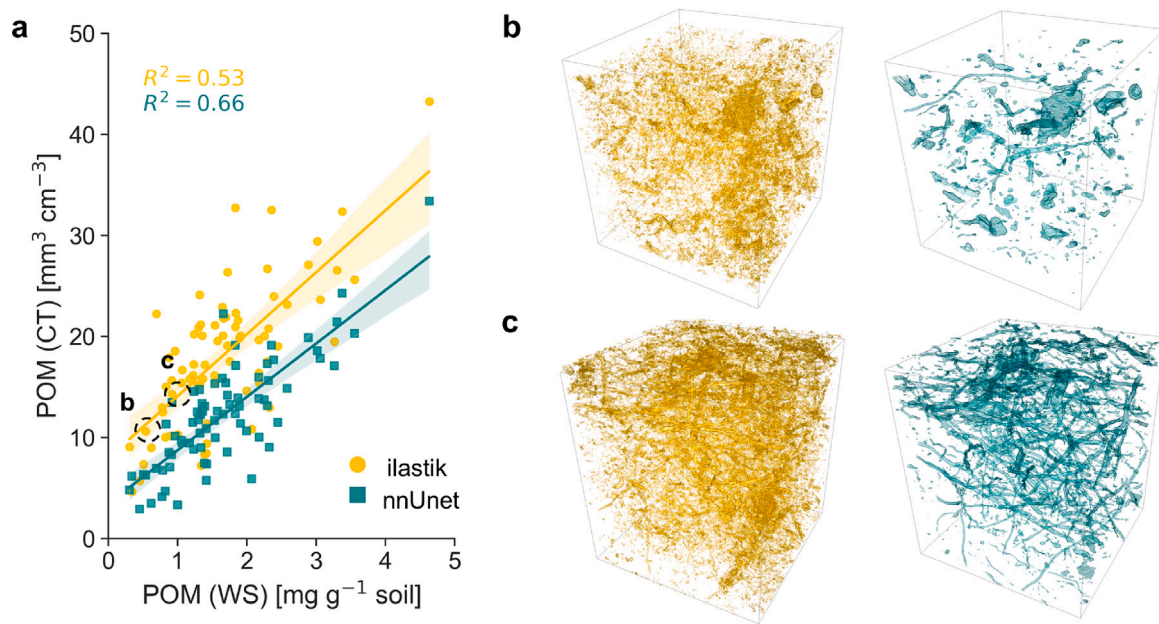


Fig. 4. Comparison of the volumetric recovery of particulate organic matter (POM) with nnUNet (in turquoise) and Ilastik (in yellow) (CT) with the recovered gravimetric POM content obtained after wet-sieving (WS) (external validation) (subfigure a). Pairwise visualization of the recovery of POM obtained with Ilastik (left-hand side, depicted in yellow) and nnUNet (right-hand side, depicted in turquoise). The letters *b* and *c* denote the samples highlighted with a dashed line circle on subfigure *a*.

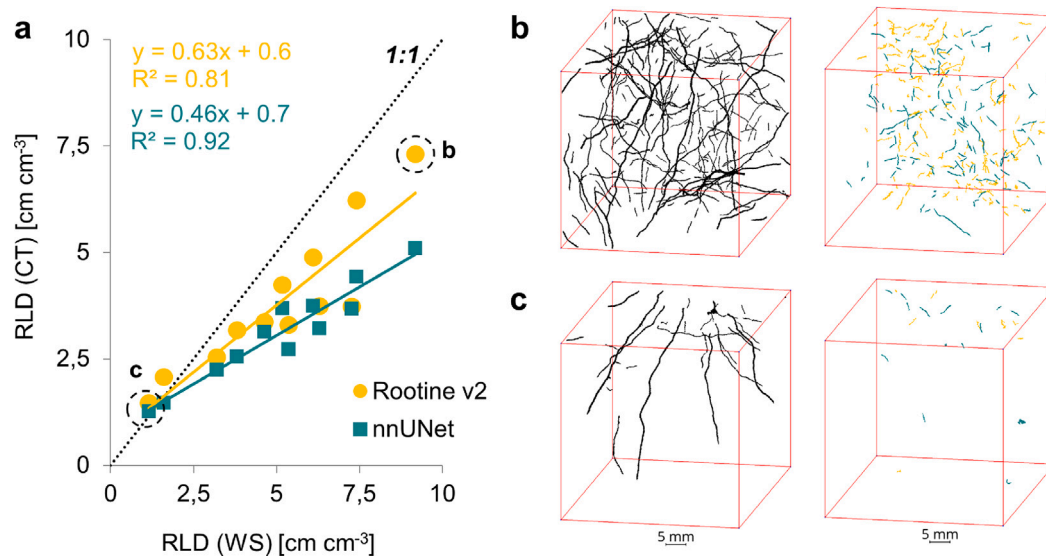


Fig. 5. Comparison of the recovery of root length density (RLD) with nnUNet (in turquoise) and Rootline v.2 (in yellow) (CT) with the recovered RLD obtained after wet-sieving (WS) (external validation) (subfigure a). Pairwise visualization of the recovery of roots obtained with Rootline v.2 and nnUNet (subfigure b and c). The roots recovered by both algorithms are shown in black (left-hand side). The roots recovered by Rootline v.2 only are shown in yellow whereas the roots recovered by nnUNet only are shown in turquoise (right-hand side). The letters *b* and *c* denote the samples highlighted with a dashed line circle on subfigure *a*.

have low X-ray CT attenuation, which resulted in an overlapping gray value range. To better discriminate between the two, more annotations that focus on these two constituents specifically would likely have helped refine the predictions. For all other classes, the Dice scores exceeded 0.8 which can be considered an excellent model performance. The highest Dice scores were obtained for classes having narrow and characteristic gray value ranges such as the wall, rocks, and living roots. This reflects the fact that the original gray value intensity was important for class detection and that other features (e.g., edge, shape, textures) contributed less.

To our knowledge, the studies of [Valdes-Korovkin et al. \(2024\)](#) and [Rippner et al. \(2022\)](#) are the only studies who considered up

to three classes in their segmentation scheme and who reported Dice scores. For Dataset 1, the Other POM class effectively corresponds to the plant residues in the study of [Valdes-Korovkin et al. \(2024\)](#) and to the organic matter investigated by [Rippner et al. \(2022\)](#). Those studies reported Dice scores of 0.57 and 0.75 for these classes. In comparison, our model achieved a Dice score of 0.8. For the abiotic pores, we have obtained a Dice score of 0.86, which is approximately equal to the Dice score for the pore class obtained by [Valdes-Korovkin et al. \(2024\)](#) (0.87), and is greater than the Dice score reported by [Rippner et al. \(2022\)](#) (0.76). For the mineral phase, the Dice score obtained with nnUNet reached 0.87, which is somewhat lower than the Dice scores reported by [Valdes-Korovkin et al. \(2024\)](#) (0.97) and [Rippner](#)

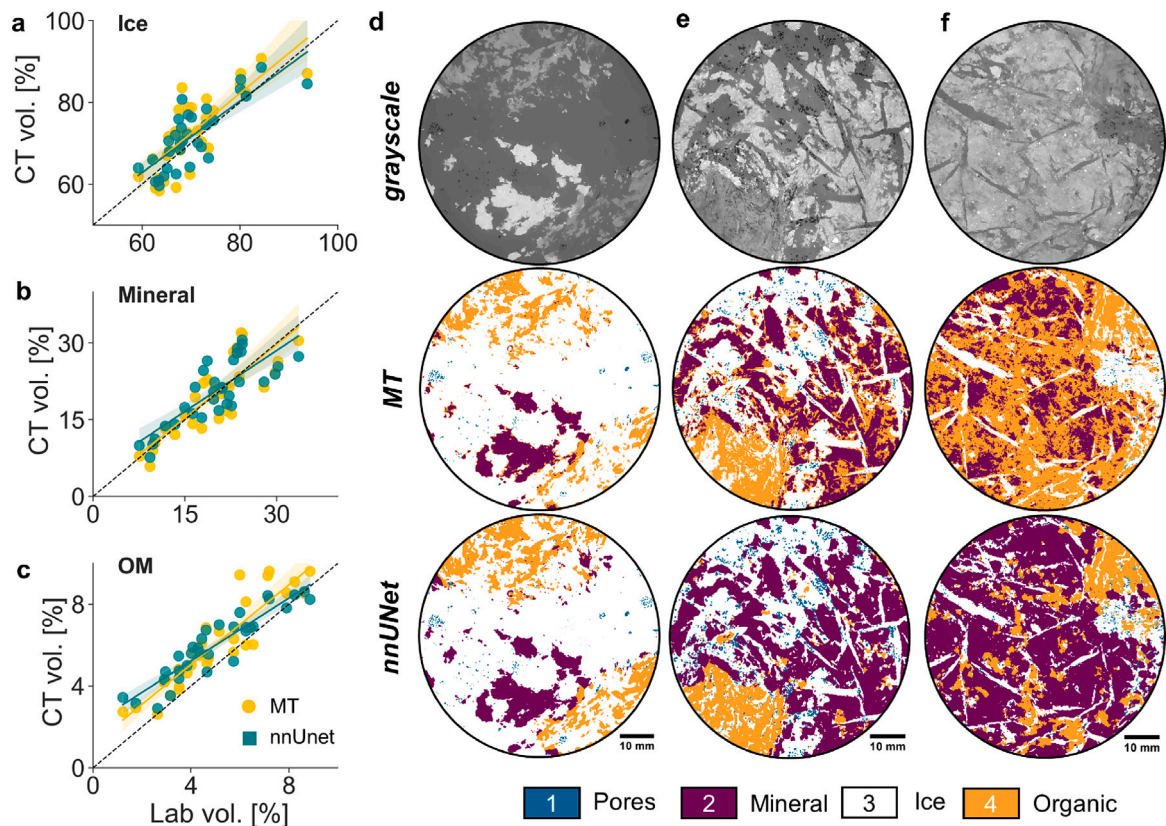


Fig. 6. Comparison of the recovery of ice (subfigure *a*), mineral (subfigure *b*) and organic matter (OM) (subfigure *c*) with nnUnet (in turquoise) and manual thresholding (in yellow) (MT) with the laboratory results obtained after thawing the samples (external validation). Visual comparison of 2D cross-sections of original grayscale images (top) with the segmented results obtained with MT (middle) and nnUnet (bottom) (subfigure *d*, *e*, *f*).

Table 4
Parameters of the linear regressions calculated between the laboratory recovered volumetric fractions of ice, mineral and organic matter and the segmented results obtained with nnUnet and manual thresholding (MT). The letters *a*, *b* indicate the slope and the intercept of the linear regressions, respectively.

Classes	MT			nnUnet		
	<i>a</i>	<i>b</i>	R ²	<i>a</i>	<i>b</i>	R ²
Ice	0.99	2.68	0.58	0.87	10.32	0.63
Mineral	0.98	0.47	0.65	0.79	4.67	0.63
Organic matter	0.96	1.18	0.83	0.76	2.15	0.84

et al. (2022) (0.91). These results demonstrate that nnUnet performed comparably well to the CNNs used in the studies by Valdes-Korovkin et al. (2024) and Rippner et al. (2022), despite being trained on a far more complex dataset. Although those studies segmented three relatively distinct classes, Dataset 1 contained up to nine classes with overlapping gray value ranges.

To compare the Dice scores obtained for Dataset 2, we refer to the studies of Soltaninejad et al. (2020) and Teramoto and Uga (2024). In a study with rice roots grown in calcinated clay and scanned with X-ray CT with varying number of projections and scan duration, Teramoto and Uga (2024) obtained Dice scores ranging from 0.46 (short scan duration) to 0.58 (long scan duration) after segmentation with a deep learning model named “VoxResNet”. Soltaninejad et al. (2020) implemented a volumetric UNet (equivalent to nnUnet) to segment intact wheat roots grown in remoulded soil scanned with X-ray CT. They reported a Dice score of 0.50. In our study, the Dice score for the living roots of Dataset 2 was equal to 0.83. This number illustrates that nnUnet was able to capture well the feature of the roots and that it provided segmentation results which were in good accordance with the annotated ground-truth data.

Compared to the reference method Routine v.2, nnUnet exhibited a lower recovery of fine roots in Dataset 2, but a greater error consistency. This can be attributed to the significant noise in the images and the fine diameter of the maize roots, which were close to the resolution limit. This challenge hindered an accurate annotation of the fine roots, resulting in their omission during the training process. With a deterministic method such as Routine v.2, root recovery can be modulated by adjusting key parameters such as the user-defined gray value range for roots or the vesselness threshold during the false negative recovery step (see figure 7 in Phalempin et al., 2021). By fine-tuning these parameters, root recovery can be enhanced, albeit sometimes at the cost of precision, as an increase in false positives might occur. Comparing visually the results of Routine v.2 and nnUnet for an image containing many roots (Fig. 5b), it can be debated whether the recovery by Routine v.2 consisted of real root segments or whether these were small pore fragments that were detected as roots (false positives). It can also be argued that the roots recovered by nnUnet seem somewhat more continuous and elongated than those recovered with Routine v.2 (Fig. 5b), a feature that is more evocative of real roots as opposed to short and bulky false positives. To improve root recovery with nnUnet, we could have modified the default softmax probabilities averaging scheme. This adjustment would have introduced a tunable parameter, making it easier to modulate root recovery, although with the risk of incorporating false positives into the segmented results. We chose not to implement this modification because we aimed to test the versatility of nnUnet in its original form.

To our knowledge, few studies attempted to unravel the constitution of permafrost soils with X-ray CT (Romanenko et al., 2017; Blais et al., 2025; Rooney et al., 2022). However, none of these studies attempted to segment the constituents of the permafrost with a deep learning method, making the comparison of the Dice scores impossible. Therefore, only a comparison with the reference segmentation method

is possible. Visual inspection of the segmented images revealed that the boundaries (edges) between the constituents were better defined with nnUNet. This can be attributed to the fact that annotations at the boundary between classes were done on larger regions of the image, which effectively smoothed out the partial volume voxels. In contrast, manual thresholding methods are prone to classification errors at the boundary between constituents because thresholds are applied on each voxel individually and disregard spatial context. Performing a visual comparison of the results of nnUNet and MT, we observed different delineations of organic matter and mineral phase (Fig. 6f). This discrepancy can be attributed to the subjectivity inherent in annotating classes with poorly defined boundaries. In addition, both classes were by nature a mixture of mineral, organic matter, and ice at the resolution employed. In such cases, we acknowledge that this segmentation challenge was ill-posed and the difficulty in properly annotating these regions contributed to the poorer results, in comparison to the volumetric fractions obtained after destructive sampling.

4.2. Applicability and practicability of nnUNet

Based on our experience, we assert that nnUNet is a powerful and versatile tool for semantic segmentation tasks applied to soil sciences. Across three datasets, nnUNet demonstrated strong performance compared to methods specifically developed for certain segmentation tasks, such as Rootline v.2, or labor-intensive approaches such as detailed manual thresholding. One of the strengths of nnUNet is its support for sparse annotations, which is particularly valuable in soil science, where annotating irregular or rare features within a dataset is often necessary. This flexibility makes nnUNet well suited for a wide range of segmentation tasks in soil sciences.

However, a limitation of deep learning models such as nnUNet became apparent during the generation of ground-truth data, as image annotation remains a laborious step. Despite employing sparse annotations (i.e., less than 0.26% of voxels were annotated), we estimated that annotating the images took approximately 2 to 4 h for each of the 20 substacks of Dataset 1. In some cases, iterative refinement of the annotations was necessary, further increasing the time investment for annotations to several days per dataset. In comparison, image annotation for random forest classifiers, as implemented in tools such as Ilastik or Labkit, typically requires fewer annotations. Training the Ilastik classifier applied in Dataset 1 required approximately 4 h only.

Given this limitation, we acknowledge that the application of deep learning models such as nnUNet for image processing in soil science may currently be best suited for application on large datasets, for which annotation efforts can be well rewarded by an increase in segmentation accuracy. A good example would be datasets, for which several hundreds of samples are extracted from the same study site (Phalempin et al., 2025; Lucas et al., 2019). Another example would be datasets generated in root phenotyping facilities, for which hundreds of growing pots are filled with the same soil material but planted with varying plant genotypes, to study root traits in an automated fashion. Dataset 1 and 2 were focused on these two potential applications for nnUNet.

To promote a larger adoption of nnUNet within the soil and plant science communities, we have made the code freely available (<https://github.com/MaxPhal/nnUNet4SoilXrayCT>), detailing the steps necessary to transition from grayscale X-ray CT images to nnUNet predictions, including image annotation. This open resource aims to streamline the application of deep learning in soil analysis with X-ray CT, making it more accessible and encouraging its adoption by the soil science community.

4.3. Outlook

Our comprehensive analysis and segmentation of complex soil structures focused on features that display a high degree of morphological consistency under various environmental conditions. For example,

roots and root channels in soils maintain a cylindrical shape regardless of depth, texture, or climatic zone. These distinguishable traits indicate that deep learning models such as nnUNet hold significant promise for the development of a generalized semantic segmentation model. Using a sufficiently diverse and representative dataset, such a model will be trained to generalize effectively across various soil conditions. The next logical step in this research is to train such a generalized model for soil structures commonly encountered in agroecosystems, including croplands, grasslands, and natural vegetation. To achieve this, the model will be trained on images from mineral soils that represent a wide spectrum of soil textures, from sand to clay-rich soils. The model will also be trained with images of different quality, that is, different image resolutions and noise levels. This will enhance the robustness and adaptability of the model, making it applicable to a wide range of soil conditions and image acquisition modalities.

In recent work, Weller et al. (2022) introduced a database infrastructure with a standardized meta-information and evaluation of segmented images. A true standardization of X-ray image analyses requires also a standardized and unbiased image segmentation. Deep learning models, and nnUNet in particular, now offer the potential to further standardize the segmentation of soil samples by enabling the development of a generalized model. The long-term objective of this research is to integrate a deep learning-based model, such as nnUNet, for the semantic segmentation of soil images uploaded to the open Soil Structure Library (Weller et al., 2022). We envision that coupling the Soil Structure Library with a robust, generalized segmentation model will be a significant step forward in fully standardizing X-ray CT image analysis for application in soil sciences. Such a framework would not only enhance the reproducibility of soil structure studies but also facilitate more precise automated analyses. This, in turn, could lead to improved soil monitoring and management practices by providing researchers and land managers with a powerful tool for assessing soil structural indicators across different agro-ecosystems.

5. Conclusion

In this study, we demonstrated the potential of nnUNet, a deep learning-based semantic segmentation method originally designed for biomedical imaging, for analyzing X-ray CT images of complex soil structures. nnUNet proved effective in segmenting multiple classes with overlapping gray value ranges, showcasing its ability to handle intricate features of the soil structure. However, we also identified certain limitations related to image annotation, particularly in challenging datasets. For instance, in a dataset of fine roots in noisy images, we faced difficulties in annotating fine roots that were close to the resolution limit. Similarly, annotating classes with gradual transitions instead of sharp boundaries such as the mineral phase and organic matter in Dataset 3 was a subjective process. Ultimately, image annotation remains an observer-dependent process, subject to variability due to differences in individual interpretation, expertise, and annotation criteria.

These findings underscore that the accuracy of predictions of deep learning models such as nnUNet is highly dependent on the quality and precision of the training data. Despite using a new approach that supports sparse annotations, the generation of ground-truth data remained time-consuming. However, the strong performance of nnUNet in a dataset with complex soil structure suggested that, with further development and access to more diverse and well-curated annotations, deep learning models such as nnUNet could help develop a generalized segmentation model, thus playing a transformative role in standardizing soil structure analysis. This could pave the way for more automated and precise monitoring of soil structural indicators, offering significant benefits to agricultural practices and environmental research.

CRediT authorship contribution statement

Maxime Phalempin: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lars Krämer:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maik Geers-Lucas:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Fabian Isensee:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Steffen Schlüter:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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